A Proposal for a Mach-Zehnder Interferometer (MZI)

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***Abstract*—This proposal gives a high level technical description of the function of an MZI and how the design and theory will drive the results from the experimental data from fabricating the circuit.**

***Index Terms*— Silicon Photonics, Mach-Zehnder Interferometer,**

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I. INTRODUCTION

THE Mach-Zehnder Interferometer is a passive device that couples light into diverging waveguides of different path lengths that when measured can give insight into the constructive and destructive nature of light. This study into the nature of light can allow for measurement of relative phase shift to further extract performance of the device such as free spectral range (FSR) and group velocity and once the device is fabricated conduct an analysis in the difference in results.

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II. THEORY

The function of the MZI is to bring in light from a source coupled from a Bragg grating coupler then split into bifurcating y branches that split the light and power into diverging branches that converge back into the same incident power and out a Bragg grating coupler for detection.
This behavior can be modeled mathematically as follows. Equation 1 represents the incident power broken into the constructive halves that diverge into the waveguides

$$\frac{E_i}{\sqrt{2}} = E_1 = E_2$$

At the out put the electric fields will converge back and take the general form of

$$E_o = \frac{E_1 + E_2}{\sqrt{2}}$$

Since this device is chosen to study the relative phase shift with respect to the source the equation will take a form of a plane wave to account for measuring and calculating phase shifts at the output for the respective path lengths. Thus the output branches can be found as follows

$$E_{o1} = E_1 e^{-i\beta L_1} = \frac{E_i}{\sqrt{2}} e^{-i\beta L_1}$$

$$E_{o2} = E_2 e^{-i\beta L_1} = \frac{E_i}{\sqrt{2}} e^{-i\beta L_2}$$

These terms account for the physical wave propagation of the light with the propagation constant $\beta = \frac{2\pi n}{\lambda}$ which is dependent on the wavelength of the light and the index of refraction within the waveguide.
With these equations an equation for the output in a losses can be constructed as

$$E_o = \frac{E_{o1} + E_{o2}}{\sqrt{2}} = \frac{E_i}{2}\left(e^{-i\beta L_1} + e^{-i\beta L_2}\right)$$

Likewise the output intensity of the light can be calculated using this form

$$I_o = \frac{I_i}{4}\left(e^{-i\beta L_1} + e^{-i\beta L_2}\right)^2 = \frac{I_i}{2}(1 + \cos(\beta \Delta L))$$

III. MODELING AND SIMULATION

A. *Waveguide*

The design parameters for the waveguide for this circuit are a height of 220*nm* and width of 500*nm* width. The mode profiles of these waveguides were simulated using lumerical FDTD solver and are shown in the figures below for the TE and TM modes respectively.

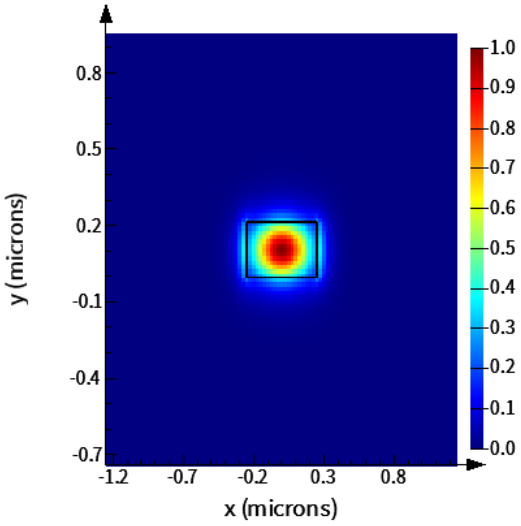


Figure 1 Electric Field Intensity of TE Mode of Waveguide

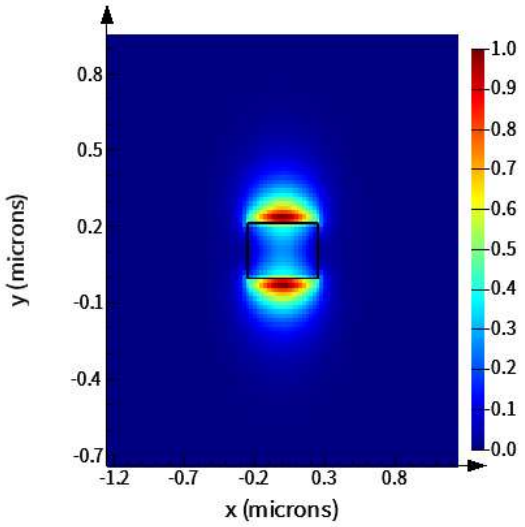
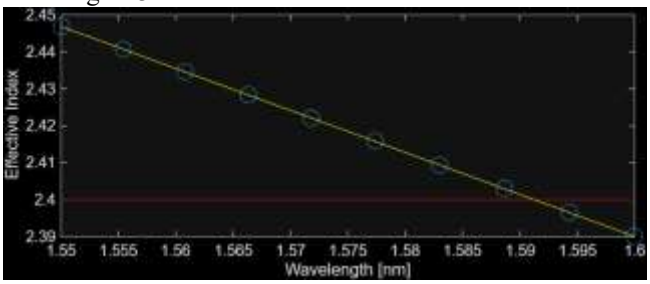


Figure 2 Electric Field Intensity of TM Mode of Waveguide

Figure 1 and Figure 2 show the electric field intensity in the transverse plane of the waveguide for the TE and TM mode respectively.
For this simulation an analysis for the group index with respect to wavelength will give insight to the waveguides performance. This is found using MATLAB with a compact model of group velocity with respect to wavelength. The equation below is used in MATLAB to give a range of the group index plotted over a range of wavelengths

$$n_{eff} = 2.44 - 1.13(\lambda - \lambda_o) - 0.03(\lambda - \lambda_o)^2$$

Using the compact model we can find the value of the effective index in Figure 3



B. Mach-Zehnder Interferometer

Now with the interferometer the transfer function must be considered with the following

$$T_{MZI} = \frac{1}{2}(1 + \cos(\beta \Delta L))$$

The free spectral range (FSR) is found using the equation below

$$FSR = \frac{\lambda^2}{(n_g \Delta L)}$$

With this equation the FSR can be calculated for the following length differences ΔL that will be used in the circuit simulation and validated in the experimental data when the chip is fabricated.

$\Delta L\ (\mu m)$	$FSR\ (nm)$
50	11.44
75	7.672
100	5.72
125	4.576
150	3.813
200	2.86
250	2.288

Table 1 ΔL vs FSR

The FSR can be validated in the simulation by using simulating the light source and detector for a MZI with a path length of 100 μm and measuring the distance between the peaks or troughs. This simulation is shown in Figure 4 below.

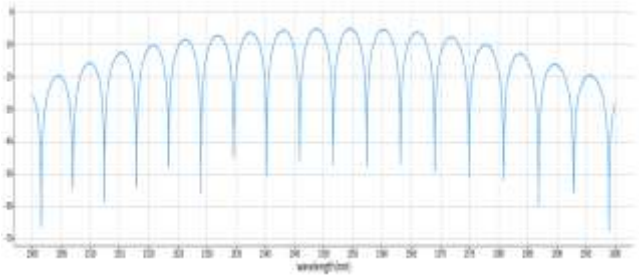


Figure 3 Simulation of MZI with ΔL of 100μm

From the figure the FSR is found to be about 6 nm which coincides with the analytical calculation.

IV. Conclusion

In this proposal the analytical approach was validated by the simulation for the MZI. These simulation and analytical results will be compared to the experimental results after fabrication.